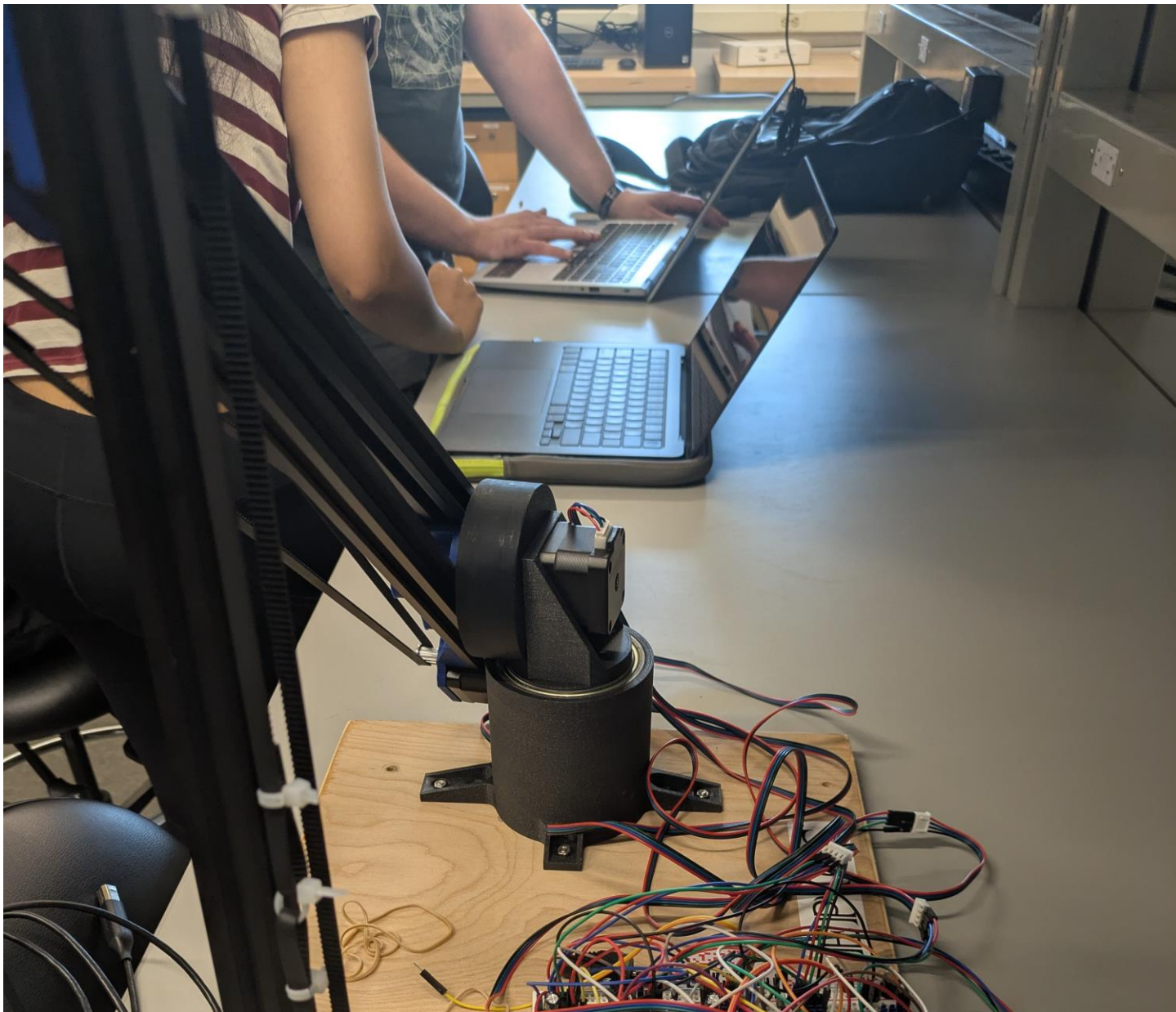




ECE Outreach

Team Manus

EPICS®

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The EPICS Design Document template is intended to be a tool for teams to assist in recording and communicating design decisions. Modifications, insertions, and deletions may be appropriate based the project discipline, scope, or other project-specific factors.

SECTION 1: PROJECT IDENTIFICATION

TUTORIALS

- Documentation: <https://tinyurl.com/EpicsDesignProcessDocument>
- Video Tutorial: <https://tinyurl.com/EpicsProjectIdentification>

PROJECT OBJECTIVE STATEMENT

- The goal of this project is to build upon the previous group's robotic arm designed for the ECE Department at Purdue. This year, Project Manus aims to create an interactive robotic arm with a claw that's fully functional and ready for presentation in front of a wide audience.

DESCRIPTION OF THE COMMUNITY PARTNER

- The ECE Outreach team aims to inform and entice younger students to learn about Purdue's Engineering department through student-led projects. To encourage them to further themselves in the field of engineering.

STAKEHOLDERS

- Prospective students who are interested in or visiting Purdue's engineering department and who aim to understand and gain accurate insight into the program would be our main potential customers other than our community partner.
- Purdue's College of Engineering Department of Electrical & Computer Engineering would gain the most from this project due to its capability of increasing interest in students, promoting them enter the program.

PROJECT SCOPE

- By the end of this project, we will have a fully functional robotic arm with a claw that can pick up items with complete housing for the circuitry that focuses on usability and aesthetics.
- When we finish our goals, we plan to leave the next team with plenty of documentation; including a guide of how to set up and operate the arm, as well as up to date files like Arduino.

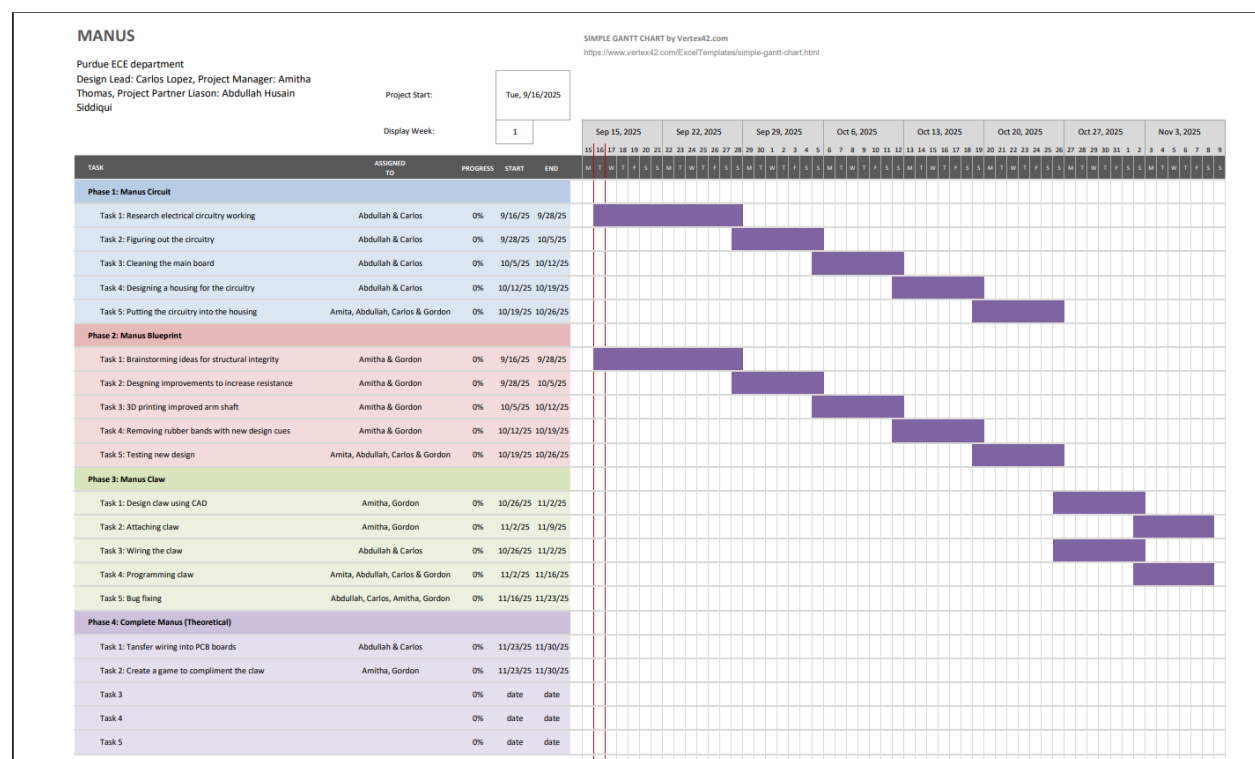
USER NEED LIST

Need #	Stakeholder	User Need
1	ECE	<i>The project must show engineering concepts to educate students</i>
2	ECE	<i>The project must fit within the budget constraints</i>
3	ECE	<i>The project must look professional and fit for display</i>
...	...	...

EXPECTED OVERALL PROJECT TIMELINE

Project Start Date: August 25, 2025

Original Target Delivery Date: April 4, 2025



- Gantt Chart Template: <https://asq.org/quality-resources/gantt-chart>

SECTION 2: SPECIFICATION DEVELOPMENT

TUTORIALS

- Documentation: <https://tinyurl.com/EpicsDesignProcessDocument>
- Video Tutorial: <https://tinyurl.com/EpicsSpecificationDevelopment>
- IP Process: <https://www.prf.org/otc/resources/commercialization.html>

DESCRIPTION OF THE USE CONTEXT

- This robotic arm will be used as an interactive display piece where students will be able to control the arm using a remote control. Since new people would be operating the system, safety measures and clear instructions would need to be provided to ensure that the people using it and the robot will remain fine.
- To prevent potential damage from elements or other incidents, the arm and its parts will be stored in lockers located in the maker space.
- The group members will be responsible for maintaining the arm and ensuring its safety.
- The technological limitations of this project include our knowledge of motion of the claw, wiring, and understanding of physics principles.
- The project will be stored indoors for the viewing and interaction of future college students. It should be in a display case at the end of the project for the poster session.

BENCHMARKING

The internet has a lot of information on building robotic arms from scratch, which would be useful to compare to.

No surveys will be taken during this project unless we require the opinion of the public for any issue, but this situation is unlikely.

All work done will be our own, including the building and the code, so no issues with copyright will occur.

SPECIFICATION LIST

Specifications are the translation of your User Needs into measurable requirements. To create a list of your project specifications, start by copying your list of user needs. For each user need, list the specifications that you must meet to satisfy that requirement. As you write your specifications, keep in mind you must be able to test the product to ensure that the specification has been met.

Need #	User Need	Spec #	Specification
1	<i>The project must display engineering techniques</i>	1.1	<i>Must document all and any ideas, plans, and progress. Make sure to address any and all concerns from our project partners.</i>
2	<i>Must be within budget restraints</i>	2.1	<i>The project must be affordable and must fit in with the requested budget.</i>
3	<i>Must look professional and fit for display</i>	3.1	<i>Must be of decent complexity to ensure that all students would be impressed by it</i>
...	...	...	...

SECTION 3: CONCEPTUAL DESIGN

TUTORIALS

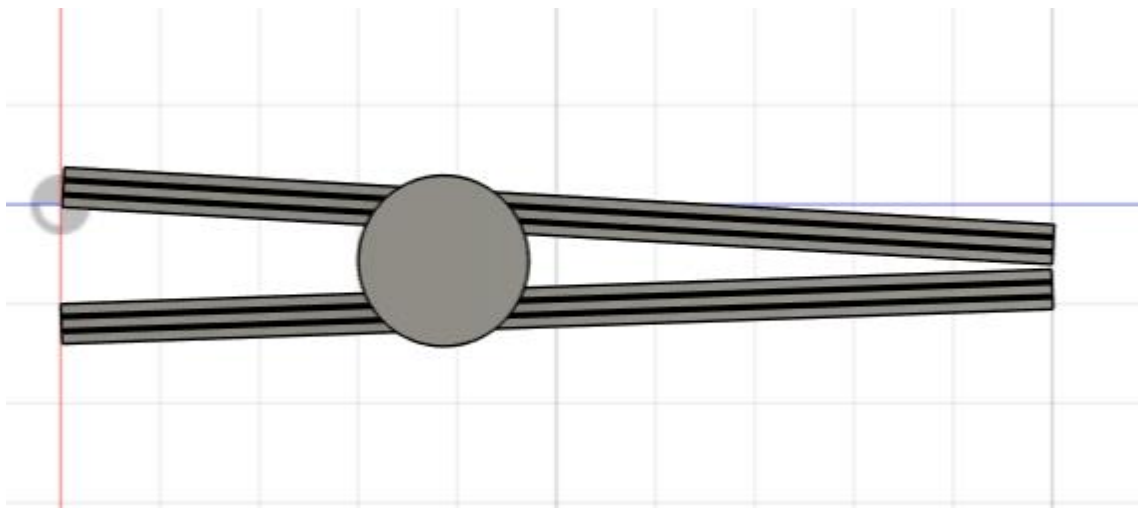
- Documentation: <https://tinyurl.com/EpicsDesignProcessDocument>
- Video Tutorial: <https://tinyurl.com/EpicsConceptualDesign>

CONCEPT GENERATION

Generating Ideas

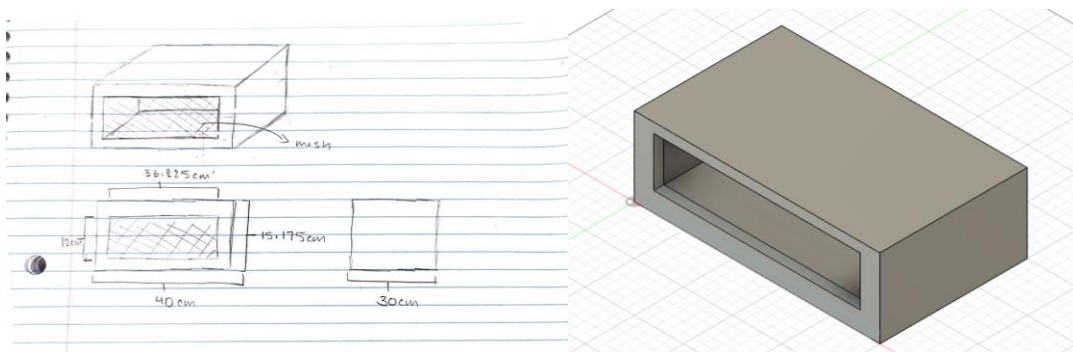
- To generate ideas, the team spent time outside of class brainstorming potential solutions. The ideas listed below were the most viable of the ones proposed.

Brainstorming to Reduce Joint Stress



- Making the arms longer while moving the pivot closer to the center was a proposed idea to help improve the counterweight so that the motor could better support the weight of the arm.
- Add a counterweight to the base of the arm (near the motor).
- Using a pulley system to reduce joint stress.

Brainstorming to Arrange Circuitry



- Housing
- Making a box with a space for a vent that can hold the circuitry with room for fans and a vent.
- Attach the vent to the box using Velcro to allow visibility of the circuitry.
- Attach the vent to the box using magnets to allow visibility of the circuitry.

This section is a place to capture artifacts (pictures, video, drawings, descriptions, etc.) from brainstorming activities. Documenting all of your viable ideas here will provide a place to come back if the solution you choose does not work out. Consider:

- What method was used to generate ideas?
- Were a sufficient number of concepts generated?
- What are the viable concepts that the team ideated?

PROTOTYPING

This section should document the early prototypes that the team has created to represent solutions. This process is iterative and may include several rounds of prototyping. Include any artifacts (pictures, videos, etc.) for reference. Consider:

- What was the purpose of the prototype (feasibility, size/aesthetic demonstration, conveying concept, etc.)?
- Was the concept intended for internal team use or for external partners?
- What simplifications were made (material, geometry, etc.)?
- How was the prototype fabricated?

As you test these prototypes internally or discuss with partners, record observations.

- What did the team learn about that concept?
- What excited your user? What frustrated them?
- What decisions or changes to the design were made based on the results?

CONCEPT CONVERGENCE

This section should include a record of the process used to select the concept(s) that the team will go forward with in development. This can be done informally through group discussion, or formally through a decision matrix or other tool. Decisions can be made for the entire concept or on a feature-by-feature basis. Scoring for concepts should be supported by prototype testing, analysis, partner or expert feedback, or other supporting evidence. Provide evidence (screen shots, whiteboard images, etc.) of the decision making process here.

- Decision Matrix Detail: <https://asq.org/quality-resources/decision-matrix>

Material Selection	Plastic			Stainless Steel		Stainless Steel + Rubber Grip	
Criteria	Weight	Score	Total	Score	Total	Score	Total
<i>Must show no wear after 50 standard wash cycles</i>	4	2	8	5	20	4	16
<i>Must be able to withstand 5 in-lb load after 50 standard wash cycles</i>	5	2	10	5	25	4	20
<i>Comfortable Handle Material</i>	2	3	6	2	4	5	10
<i>Low Cost Fabrication</i>	3	5	15	3	9	2	6
<i>Low Fabrication Lead Time</i>	2	2	4	4	8	3	6
Total			43		66		58

PROPOSED SOLUTION

This section should include a complete description of the proposed design concept, including sketches, process diagrams, or other artifacts to convey the concept. Consider whether this solution may be patentable. This solution should be approved by the advisor and partner before proceeding to detailed design.

SECTION 4: DETAILED DESIGN

REFERENCES:

- Documentation: <https://tinyurl.com/EpicsDesignProcessDocument>
- Video Tutorial: <https://tinyurl.com/EpicsDetailedDesign>

BILL OF MATERIAL (B.O.M)

This section should include a list of all of the components, whether manufactured or purchased, that go into the final design.

Sub-Assembly	Item	Catalog/ Part No.	Manufactured/ Purchased	Vendor/ Method	Quantity	Cost/ Unit
Housing	<i>Axial Fans</i>	<i>n/a</i>	<i>Pending</i>	<i>Zoro</i>	<i>3</i>	<i>\$51.87</i>
Housing	<i>Plexiglass</i>	<i>n/a</i>	<i>Pending</i>	<i>Amazon</i>	<i>2</i>	<i>\$38.99</i>
Housing	<i>Wire Mesh</i>	<i>n/a</i>	<i>Pending</i>	<i>Amazon</i>	<i>1</i>	<i>\$9.99</i>
						<i>\$12.99</i>
Robot Arm	<i>GT2 Timing Belt</i>	<i>n/a</i>	<i>Pending</i>	<i>Amazon</i>	<i>1</i>	
						<i>\$42.49</i>
Robot Arm	<i>400 mm Aluminum Extrusion</i>	<i>n/a</i>	<i>pending</i>	<i>Walmart</i>	<i>4</i>	

PRINTS/SCHEMATICS/CODE

This section should provide links to any relevant material that describe the technical details of each component and the overall assembly or technology stack, whether the project is physical, virtual, or both. These materials should be of quality and completeness such that a 3rd party of reasonable skill could fabricate the project without additional instruction.

MANUFACTURING AND ASSEMBLY PROCESSES

This section should describe the methods used to manufacture and assemble the final product. Create sub-sections as needed when major iterations of the design occur. The description should be sufficient for an average freshman engineering student to replicate the product. Consider what improvements to the process could be made and give specific recommendations for future teams.

RISK ANALYSIS

This section should include a link or pasted image of any risk analysis or risk mitigation activities. One of the most common tools for risk analysis is the Failure Mode and Effect Analysis (FMEA).

- FMEA Template: [American Society for Quality FMEA Template](#)

VERIFICATION

This section should include a table summarizing the results of verification activities for the project. A link to the verification test report should also be included in the table. Verification is the process of making sure the design outputs meet the design specifications. Each specification should be verified, which can take any form that confirms that the specification is met. Be sure to consider any residual high risks from the assessment.

Spec #	Specification	Verification
1.1	Must hold 0.75 fluid ounces of water	Volume sample analysis resulted in an average 0.83 oz capacity
2.1	Handle length between 3.5 and 5 inches	Print specifies handle length as 4 +/- 0.1 in
2.2	Handle cross sectional area < 0.1 in ²	Print specifies handle length as 0.8 +0.01/-0.02 in
3.1	Must show no wear after 50 standard wash cycles	Dishwasher fatigue test revealed no wear after 50 cycles
3.2	Must be able to withstand 5 in-lb load after 50 standard wash cycles	Dishwasher fatigue test results showed no mechanical failure under 5 in-lb load after 50 cycles
...	...	...

VALIDATION

This section should include a table summarizing the results of validation activities for the project. A link to the validation test report should also be included in the table. Validation is the process of demonstrating that the final design meets the user needs. Each user need should have at least one validation, which can take any form that confirms that the user need is met.

Need #	User Need	Validation
1	<i>Must hold one bite of liquid</i>	<i>User testing with 8 children from age 2-7 confirmed the spoon held an appropriate bite of liquid food</i>
2	<i>Must have a graspable handle</i>	<i>User testing with 8 children from age 2-7 confirmed the spoon handle was easily graspable for children above 4 years old. The product manual will be updated to reflect this limitation and the project partner has been informed.</i>
3	<i>Must be dishwasher safe</i>	<i>Dishwasher fatigue testing confirmed a dishwasher does not cause mechanical damage to the device.</i> <i>A literature search on the materials confirmed dishwashing does not induce a chemical/biological hazard.</i>
...	...	...

SECTION 5: PROJECT DELIVERY

TUTORIALS

- Delivery Process:
 - <https://engineering.purdue.edu/EPICS/teams/team-documents/project-delivery>
- Delivery Checklist:
 - <https://engineering.purdue.edu/EPICS/teams/team-documents/delivery-checklist>

**Partner agreements mandate the completion of the delivery checklist.
Failure to complete the checklist and receive EPICS administrative
approval may result in personal liability.**

**Do NOT deliver a project until the checklist is completed and
approved by both the advisor and EPICS administration.**

USER/SERVICE MANUAL

A user manual should be provided to the community partner to aid them in use and basic maintenance of the product. An in-house manual or engineer to engineer guide may also be created to aid future teams in servicing and troubleshooting the product. Insert a link to each manual here.

DELIVERY CHECKLIST

Should be completed by the team prior to delivery. Link or copy the completed checklist onto this page.

CUSTOMER SATISFACTION QUESTIONNAIRE

Should be completed by the community partner two weeks after delivery. Link or copy the completed questionnaire onto this page.

- Customer Satisfaction Questionnaire: <https://tinyurl.com/EpicsCustomerSatisfaction>

RECORD OF PROJECT DELIVERY

Please add or link to a photo and/or video of the project at the time of delivery.

SECTION 6: CURRENT SEMESTER RECORD

This section should contain information on the current semester only. It should be moved to Appendix A at the conclusion of the semester.

POINT OF CONTACT FOR FUTURE TEAM MEMBERS (E.G DESIGN LEAD)

Name:

Email:

Phone:

POINT OF CONTACT AT THE COMMUNITY PARTNER ORGANIZATION

Name:

Email:

Phone:

CURRENT PROJECT STATUS

The robotic arm from the previous team is still intact, but we are not sure if all of the wiring is in the right places.

CURRENT SEMESTER PROJECT TIMELINE

This section should address such questions as:

- What major milestones will we complete this semester?
- What details should we be working on now?
- The most common tool for project planning in industry is the Gantt Chart

TRANSITION REPORT

This section should include:

- Storage location and login info for all team materials (code, CAD models, etc)
- Major milestones completed
- Major roadblocks encountered and suggested remedy
- Suggested next-steps for next semester's team
- Team leadership roles that have been established for next semester

APPENDIX A: PAST SEMESTER RECORDS